

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Team ID	SWTID-2026-5454
Project Name	Streamlining IT Procurement – Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

1. Technical Architecture

The solution is implemented on the ServiceNow platform. The employee/IT user interacts with the Service Catalog to request a Standard Laptop. The request is approved, after which Flow Designer automatically creates a Catalog Task and assigns it to the Hardware group.

EMPLOYEE / IT USER

↓ Standard Laptop Request

SERVICENOW SERVICE CATALOG

↓ Request for Approval

APPROVER

↓ Approved Request

FLOW DESIGNER → AUTOMATED CATALOG TASK

Short Description: Laptop need to Configured | Assignment Group: Hardware

HARDWARE TEAM → LAPTOP CONFIGURATION

Infrastructure demarcation: The project is a ServiceNow platform workflow. No separate local application server, cloud database, file-storage service, external API, or machine-learning model is specified in the supplied project documentation.

2. Table-1: Components & Technologies

The following stack is limited to the components evidenced by the project implementation and the supplied template.

S.No	Component	Description	Technology
1	User Interface	Interface through which the user accesses the Standard Laptop request and order process.	ServiceNow Service Catalog
2	Application Logic-1	Logic that triggers automation when the Service Catalog process reaches the required approval stage.	ServiceNow Flow Designer
3	Application Logic-2	Logic that creates the required Catalog Task and populates its fields.	Flow Designer – Create Catalog Task Action
4	Application Logic-3	Logic for assigning the generated task to the team responsible for laptop configuration.	Flow Designer – Assignment Group: Hardware
5	Database	Stores ServiceNow request, Requested Item, approval, and Catalog Task records.	ServiceNow platform data tables
6	Cloud Database	No separate cloud database service is specified in the project documentation.	Not separately specified
7	File Storage	No separate file-storage component is required or specified for this workflow.	Not specified

8	External API-1	No external API is used or specified in the implemented workflow.	Not applicable
9	External API-2	No second external API is used or specified in the implemented workflow.	Not applicable
10	Machine Learning Model	No machine-learning model is required for the Standard Laptop procurement automation.	Not applicable
11	Infrastructure (Server / Cloud)	The workflow is executed within the ServiceNow platform environment.	ServiceNow platform / cloud-hosted ServiceNow instance

3. Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source framework is specified in the project implementation. The solution uses ServiceNow platform capabilities.	ServiceNow Flow Designer / Service Catalog
2	Security	Access to ServiceNow records and workflow actions is controlled through the platform's user roles and access controls. The flow is configured to run as a System user in the supplied implementation.	ServiceNow roles / ACLs / System User execution
3	Scalable Architecture	The automation is workflow-based and can be reused for the Standard Laptop catalog process without creating a separate application server.	ServiceNow Flow Designer
4	Availability	The workflow is available through the ServiceNow platform environment, allowing the catalog request and automated processing to be accessed through the platform.	ServiceNow platform
5	Performance	Automation reduces manual task creation and assignment steps, allowing the Catalog Task to be generated and routed after approval.	Flow Designer automation

References from template: C4 Model, IBM Cloud Architecture, AWS Architecture, and the technical architecture references listed in the supplied template.

Note: The supplied project documentation does not specify a separate frontend framework, programming language, external API, cloud database, file-storage service, or machine-learning model. Those components are therefore marked as not specified/not applicable rather than being added without evidence.